

## DISCUSSION ON THE TWO KEYS

The two keys, Key 1 ("Professional Analyst") and Key 2 ("Straightforward AI Use"), successfully model the genealogy database but differ fundamentally in their approach to efficiency, data integrity, and complexity. Key 1 represents a highly optimized, compact design focused on long-term query performance and strict business rule enforcement, while Key 2 is a more verbose, direct implementation that favors simplicity and ease of initial understanding.

The most significant divergence is in **lineage modeling**.

- Key 1 utilizes **recursive foreign keys** (Father\_ID, Mother\_ID referencing Person\_ID) directly within the People table. This is the **gold standard for graph-like data structures like genealogy**, offering the highest performance for deep lineage queries (e.g., finding all ancestors or descendants) and keeping the table count minimal.
- Key 2, in contrast, avoids the conceptual complexity of recursion **by introducing a dedicated linking table, Parentage**. While this structure is often simpler to understand for beginners, it results in an extra table and requires an extra join for every simple parent-child lookup, making it less efficient for complex historical analysis.

A second major difference lies in **data integrity via type selection**.

- Key 1 **rigorously used ENUM** data types for categorical fields like end\_reason and Acquisition\_Reason in the Titles table. This strictly limits the input values, preventing typographical errors and ensuring perfect data quality, which is vital for historical data that must remain consistent over time.
- Key 2, reflecting a straightforward implementation, used simple VARCHAR for all these fields. While flexible, this choice sacrifices integrity, as the database cannot prevent a user from entering 'death' instead of 'death (bride)' or misspelling a reason, leading to inconsistent reporting and requiring complex application-side validation.
- Furthermore, Key 1 consolidated all non-sovereign titles into a single, comprehensive Titles table, whereas Key 2 created a redundant, separate Princes table specifically for Prince/Princess titles, resulting in an unnecessary fifth table and a less unified schema.

Finally, the difference is demonstrated in the correctness of the initial composite keys.

- Key 1 demonstrated expertise by proposing the minimal and optimal **recursive foreign key** solution from the start.
- The design process for Key 2, however, required an iterative **correction** of the Parentage table's composite key; it was initially over-defined as (Child\_ID, Parent\_ID, Relationship\_Type) and had to be simplified to the minimal and correct key (Child\_ID, Relationship\_Type).

This need for correction, even on a basic primary key, contrasts with the Key 1 approach, which anticipates and avoids structural inefficiencies, highlighting the difference between a direct AI output and a carefully validated professional design.